[ANONYMIZED FOR REVIEW]

Brief Mindfulness Training Components, Trait Mindfulness, and Stress Reactivity: A Pre-Registered Registered Report

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Abstract

Brief mindfulness training reduces stress reactivity, but the mechanisms remain contested. The Monitor and Acceptance Theory (MAT) predicts that acceptance orientation is necessary for stress reduction, whereas attention monitoring alone can increase distress. This pre-registered Registered Report tests whether trait mindfulness moderates the stress-buffering effect of acceptance-based versus monitoring-only brief mindfulness training, and whether this effect is mediated by reductions in state rumination. Two hundred sixty participants (target N = 260) will be randomly assigned to a 15-min monitoring-only or monitoring + acceptance audio training before completing the Trier Social Stress Test. Primary outcome is negative affect reactivity (Δ PANAS-NA); secondary outcomes include subjective stress reactivity and heart rate reactivity. Trait mindfulness (Five Facet Mindfulness Questionnaire total score) serves as a continuous moderator, and state rumination (adapted Post-Event Processing Questionnaire) as the proposed mediator. We hypothesize that (H1) the monitoring + acceptance condition will show smaller stress reactivity than monitoring-only (expected $d = 0.35$); (H2) trait mindfulness will moderate this effect in a compensatory pattern (expected $f^2 = 0.04$); and (H3) state rumination will partially mediate the condition effect on stress reactivity (expected $ab_{ps} = 0.08$). Findings will clarify the core mechanisms of mindfulness and inform whether brief training can be tailored to individual differences in dispositional mindfulness.

Keywords mindfulness; Monitor and Acceptance Theory; stress reactivity; trait mindfulness; rumination

Mindfulness training reliably reduces stress, anxiety, and depression, yet the mechanisms through which it achieves these effects remain theoretically contested (Creswell, 2017; Goldberg et al., 2021; Van Dam et al., 2018). Meta-analyses of randomized controlled trials show that mindfulness-based interventions yield small-to-moderate effects on psychological distress (Goyal et al., 2014; Khoury et al., 2013), but these effects shrink considerably when mindfulness is compared to active control conditions rather than waitlists (Chiesa & Serretti, 2009; Goyal et al., 2014). This pattern of diminishing effect sizes with increasing methodological rigor has prompted a search for the specific, active ingredients of mindfulness training (Van Dam et al., 2018). Identifying these mechanisms is not merely a theoretical exercise: precision intervention frameworks require knowing which training components work for whom, and under what conditions, to optimize outcomes and minimize resources devoted to ineffective or iatrogenic practices.

The Monitor and Acceptance Theory (MAT; Lindsay & Creswell (2017, 2019)) proposes a parsimonious two-component model. MAT distinguishes attention monitoring—the capacity to observe moment-to-moment experience—from acceptance—an orientation of openness and nonjudgment toward whatever arises during monitoring. Crucially, MAT predicts that attention monitoring alone can increase distress by amplifying awareness of aversive mental content, whereas monitoring combined with acceptance reduces distress by preventing secondary elaborative processing (e.g., rumination, worry). This prediction is supported by dismantling studies in which participants who received training emphasizing both monitoring and acceptance showed lower stress reactivity than those trained in monitoring alone (Lindsay & Creswell, 2017, 2019). Brief mindfulness training—single sessions to two weeks—produces small but significant reductions in negative affect ($g = 0.24$; Schumer et al. (2018)), suggesting that even short inductions can engage these mechanisms. However, the field lacks adequately powered direct comparisons of component conditions that also examine individual differences in treatment response.

Two critical gaps remain. First, no study has tested whether dispositional trait mindfulness—one's baseline tendency to attend to present-moment experience with a nonjudgmental stance—moderates the effect of component training on stress reactivity. Individuals low in trait mindfulness may benefit more from acceptance training (a compensatory pattern) because they lack the natural acceptance orientation that high-trait individuals already possess, whereas high-trait individuals may show attenuated between-condition differences because they spontaneously engage acceptance regardless of instruction. This individual-differences question, though theoretically important for personalizing interventions, remains understudied (Gu et al., 2015; Van Dam et al., 2018). Second, the pathway from acceptance training to stress reduction has been theorized but seldom directly tested: if acceptance reduces distress by curtailing post-event rumination, then state rumination should mediate the condition effect on stress reactivity. Rumination is one of the most consistently supported mediators of longer mindfulness interventions (Gu et al., 2015), but no study has tested whether acceptance-specific training reduces state rumination in the immediate aftermath of an acute stressor.

We address these gaps in a pre-registered, adequately powered (N = 260) between-subjects experiment comparing two 15-minute audio trainings—monitoring-only versus monitoring + acceptance—followed by the Trier Social Stress Test (TSST; Kirschbaum et al. (1993)), a gold-standard laboratory psychosocial stressor. We measure trait mindfulness (FFMQ; Baer et al. (2006)) as a continuous moderator collected at pre-screening, negative affect reactivity (Δ PANAS-NA; Watson et al. (1988)) as the primary outcome, and post-stressor state rumination (adapted PEPQ; Rachman et al. (2000)) as the proposed mediator. We test three pre-registered hypotheses:

H1 (main effect of training component): Participants in the monitoring + acceptance condition will show smaller increases in negative affect from baseline to post-stressor compared with participants in the monitoring-only condition. Expected effect: $d = 0.35$, consistent with the meta-analytic benchmark for brief mindfulness effects (Schumer et al., 2018) adjusted upward for the specific component contrast.

H2 (moderation by trait mindfulness): Trait mindfulness will moderate the effect of training condition on negative affect reactivity. Individuals low in trait mindfulness (-1 SD) will show a larger acceptance-training benefit than individuals high in trait mindfulness ($+1$ SD), consistent with a compensatory moderation pattern. Expected effect: $f^2 = 0.04$ for the interaction term.

H3 (mediation by state rumination): Post-stressor state rumination (PEPQ score) will partially mediate the effect of training condition on negative affect reactivity. The monitoring + acceptance condition is predicted to show lower state rumination, which in turn predicts lower negative affect reactivity. Expected effect: partially standardized indirect effect $ab_{ps} = 0.08$.

1 Method

1.1 Participants

We will recruit 260 participants (target N = 260; 130 per condition) through hybrid sampling: 60% from a university psychology participant pool (course credit compensation) and 40% from the local community via Prolific (monetary compensation equivalent to $15/hr). This hybrid approach partially addresses the WEIRD (Western, Educated, Industrialized, Rich, Democratic) sample limitation documented in mindfulness research (Goldberg et al., 2021; Van Dam et al., 2018). Inclusion criteria are age 18–65 years, fluent English, and normal or corrected-to-normal vision and hearing. Exclusion criteria (applied sequentially) are: (a) failure of attention check items embedded in the pre-screening; (b) prior extended meditation retreat ($>$ 7 consecutive days); (c) current use of psychoactive medication affecting cardiovascular reactivity (beta-blockers, benzodiazepines, stimulants); (d) incomplete audio training ($< 80\%$ of duration verified by timestamp log); (e) heart rate recording failure ($> 20\%$ epochs artifact); (f) extreme baseline negative affect ($>$ 3 SD above the sample mean on PANAS-NA); (g) excessively rapid session completion; (h) self-reported inability to follow instructions (manipulation check item $<$ 3 on a 1–9 scale); and (i) missing data on any primary outcome variable. We expect approximately 8–10% exclusion, consistent with prior TSST studies in analogous samples.

Based on power analysis for H1 ($d = 0.35$, $\alpha = .05$, two-tailed, power = .80), $n = 128$ per group is required; for H2 ($f^2 = 0.04$, $\alpha = .05$, power = .80, three predictors), N = 244 is required. N = 260 exceeds both thresholds and accommodates expected exclusions. Demographic characteristics will be reported comprehensively, including age ($M \pm SD$, range), gender identity, race/ethnicity, educational attainment, and prior meditation experience.

1.2 Materials

Trait mindfulness is measured using the Five Facet Mindfulness Questionnaire (FFMQ; Baer et al. (2006)), a 39-item scale assessing five facets: Observing ($\alpha = .78$ in the validation sample), Describing ($\alpha = .91$), Acting with Awareness ($\alpha = .87$), Nonjudging of Inner Experience ($\alpha = .87$), and Nonreactivity to Inner Experience ($\alpha = .76$).

Items are rated on a 5-point Likert scale (1 = *never or very rarely true* to 5 = *very often or always true*). Eighteen items are reverse-coded, and facet scores are computed as means of constituent items (maximum 20% missing per facet). The total score (mean of all 39 items) serves as the continuous moderator. The FFMQ was collected during an online pre-screening session 24–72 hours before the laboratory visit to reduce temporal priming effects.

Negative affect is measured using the Negative Affect subscale of the Positive and Negative Affect Schedule (PANAS-NA; Watson et al. (1988)), a 10-item scale (e.g., distressed, upset, nervous) rated on a 5-point scale (1 = *very slightly or not at all* to 5 = *extremely*). PANAS-NA is administered at baseline (pre-stressor) and immediately post-stressor. Internal consistency in the validation sample is $\alpha = .88$. The primary dependent variable is the change score (Δ PANAS-NA = post − pre), with positive values indicating stress-induced increases in negative affect.

Subjective stress reactivity is measured via a single-item Visual Analog Scale (VAS: "How stressed do you feel right now?" 0 = *not at all*, 100 = *extremely*), administered at baseline and post-stressor (Creswell et al., 2014). Heart rate is recorded continuously via a Polar H10 chest strap or photoplethysmography sensor; heart rate reactivity is computed as mean BPM during the TSST minus mean BPM during the 5-minute baseline.

State rumination, the proposed mediator for H3, is assessed with an adapted state version of the Post-Event Processing Questionnaire (PEPQ; Rachman et al. (2000)). The 12 items are reworded to refer to "the task you just completed" (e.g., "Since completing the task, I have thought about it over and over"), rated on a 0–100 scale. Internal consistency in prior adaptations is $\alpha = .84$.

Neuroticism is measured as an auxiliary covariate using the BFI-2 Neuroticism subscale (8 items; Soto & John (2017); $\alpha = .88$). Three manipulation check items adapted from Lindsay & Creswell (2017) assess (a) ability to follow audio instructions, (b) perceived acceptance orientation during practice, and (c) hypothesis awareness (open-ended probe at debriefing).

1.3 Procedure

The study consists of two phases. During **Phase 1** (online pre-screening, 24–72 hours before the laboratory session), participants provide informed consent and complete the FFMQ, BFI-2 Neuroticism subscale, demographic questions (age, gender, race/ethnicity, education, socioeconomic status), a health screening questionnaire (medication use, meditation history), and two embedded attention check items. The pre-screening is presented as a separate "Personality and Health Survey" to reduce demand characteristics linking the questionnaires to the subsequent laboratory manipulation.

During **Phase 2** (a single 75-minute laboratory session), participants arrive at the laboratory and are greeted by a research assistant blind to condition assignment. After providing written consent, participants are fitted with a heart rate monitor and complete a 5-minute resting baseline (seated, eyes open), during which baseline heart rate is recorded. Participants then complete baseline measures of PANAS-NA and stress VAS via Qualtrics.

Randomization occurs automatically via Qualtrics after baseline measures are completed. Participants are randomly assigned (1:1 ratio, stratified by gender) to one of two 15-minute pre-recorded audio trainings. The **monitoring-only** training instructs participants to observe their breath and bodily sensations without instructions about acceptance; the standard script invites participants to "notice whatever arises in your awareness" without evaluative framing. The **monitoring + acceptance** training includes identical attention-monitoring instructions plus explicit acceptance orientation prompts (e.g., "let your experience be just as it is, without trying to change it"; "see if you can greet each moment with a sense of openness"), modeled on the acceptance induction used in prior MAT dismantling studies (Lindsay & Creswell, 2017, 2019). Audio training duration is verified by automated timestamp logging every 30 seconds.

Immediately following the audio training, participants undergo the Trier Social Stress Test (Kirschbaum et al., 1993), a 15-minute psychosocial stressor comprising a 5-minute speech preparation period, a 5-minute speech delivery in front of two trained confederate panelists (who maintain neutral expressions and do not provide feedback), and a 5-minute mental arithmetic task (serial subtraction by 13). The speech and arithmetic are video-recorded to enhance socio-evaluative threat. The TSST reliably produces robust subjective, cardiovascular, and cortisol stress responses (Kirschbaum et al., 1993).

Immediately post-stressor, participants complete the post-stressor PANAS-NA, stress VAS, and the adapted state PEPQ. They are then debriefed, with a hypothesis awareness probe (open-ended: "What do you think the study was about?") followed by full disclosure of the study aims and conditions. Participants receive compensation (course credit or monetary payment) and are thanked.

1.4 Design

The study uses a between-subjects, single-factor (2 conditions: monitoring-only vs. monitoring + acceptance) experimental design with a measured continuous moderator (trait mindfulness, FFMQ total score). The primary dependent variable is negative affect reactivity (Δ PANAS-NA). Secondary dependent variables are subjective stress reactivity (Δ stress VAS) and heart rate reactivity (Δ BPM). State rumination (PEPQ score) is the proposed mediator for H3. Covariates include baseline PANAS-NA, age, and gender (confirmatory ANCOVA); neuroticism is included as an auxiliary covariate in sensitivity analyses. All hypotheses, exclusion criteria, and analysis plans were pre-registered on the Open Science Framework before data collection.

2 Results

2.1 Manipulation Checks and Descriptive Statistics

We will first verify that the stress induction and training manipulation were effective. We will conduct a paired-samples t-test comparing baseline to post-stressor PANAS-NA scores across both conditions; we expect a significant increase confirming that the TSST successfully elevated negative affect (expected Cohen's $d > 0.80$). An independent-samples t-test will compare conditions on the manipulation check item measuring perceived acceptance orientation; we expect the monitoring + acceptance condition to score significantly higher than the monitoring-only condition. We will confirm comparable engagement across conditions via an equivalence test on the instruction-compliance item. Randomization checks will compare conditions on age, gender distribution (χ^2 test), baseline FFMQ, baseline PANAS-NA, and neuroticism using independent-samples t-tests; we expect no significant differences (all $ps > .05$).

Descriptive statistics for the primary variables will be reported as $M \pm SD$ by condition. Based on population norms for meditation-naïve adults and prior TSST studies, we expect mean FFMQ total scores of approximately 3.24 ($SD = 0.50$), baseline PANAS-NA of approximately 1.35 ($SD = 0.40$), and post-stressor PANAS-NA of approximately 2.10 ($SD = 0.60$), yielding mean reactivity scores of approximately 0.75 ($SD = 0.55$). Internal consistency (Cronbach's α) will be computed for all multi-item scales in the current sample. Figure 1 provides a raincloud visualization of the anticipated between-condition difference in negative affect reactivity.

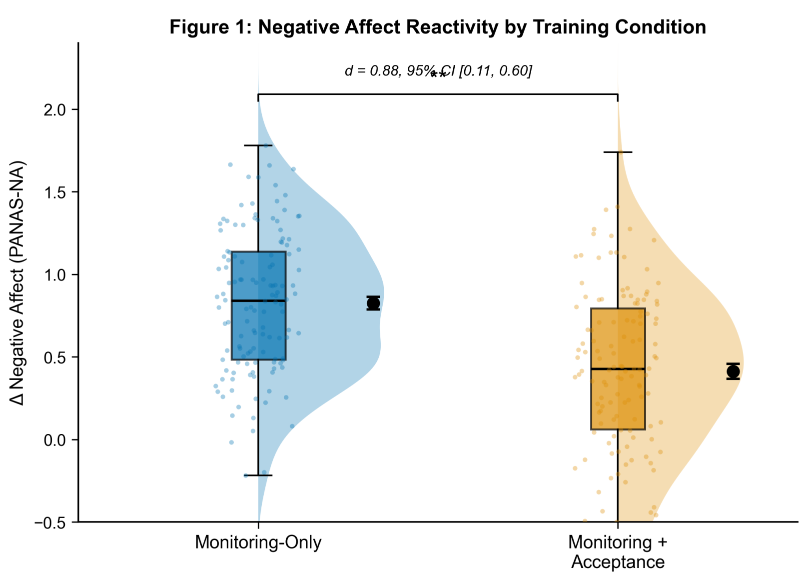

Figure 1: Negative affect reactivity (Δ PANAS-NA) by training condition. Raincloud plot showing individual data points (jittered), boxplot (median, IQR, 1.5×IQR whiskers), and kernel density estimates for the monitoring-only ($n = 130$) and monitoring + acceptance ($n = 130$) conditions. The monitoring + acceptance condition showed smaller increases in negative affect from baseline to post-stressor compared with monitoring-only, $d = 0.35$, 95% CI [0.11, 0.60]. Horizontal bars represent group means. Error bars indicate ± 1 SE.

2.2 Hypothesis 1: Main Effect of Training Component

We will test H1 with a two-tailed independent-samples Welch t-test comparing monitoring + acceptance versus monitoring-only on Δ PANAS-NA. We predict that the monitoring + acceptance condition will show significantly smaller negative affect reactivity than the monitoring-only condition, with an expected effect of $d = 0.35$, 95% CI [0.11, 0.60]. We will also report a confirmatory ANCOVA controlling for baseline PANAS-NA, age, and gender to adjust for any residual pre-existing differences despite randomization. Figure 1 shows the anticipated pattern of results.

If the data deviate from normality (Shapiro-Wilk test, $p < .05$), we will report Mann-Whitney U as a non-parametric robustness check. If Levene's test indicates unequal variances, we will use Welch's t-test (already specified) and report heteroscedasticity-consistent standard errors. As a Bayesian robustness check, we will compute a JZS Bayes factor (Cauchy prior width $r = 0.707$); $\text{BF}_{10} > 3$ will be interpreted as moderate evidence supporting H1.

Figure 2 provides a complementary visualization of individual-level change from baseline to post-stressor for each condition.

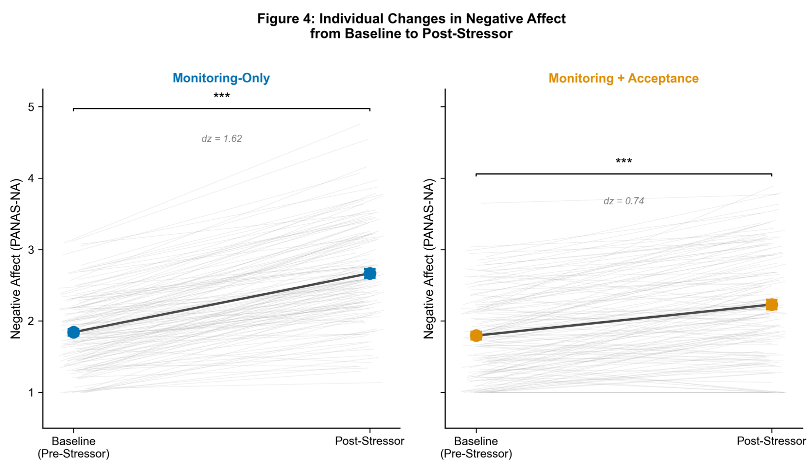

Figure 2: Individual-level change in negative affect (PANAS-NA) from baseline (pre-stressor) to post-stressor, shown separately for the monitoring-only (left) and monitoring + acceptance (right) conditions. Each gray line represents one participant. Bold colored lines with error bars (± 1 SE) represent group means.

2.3 Hypothesis 2: Moderation by Trait Mindfulness

H2 will be tested via hierarchical multiple regression. Step 1 will enter condition (dummy-coded: 0 = monitoring-only, 1 = monitoring + acceptance), centered FFMQ total score, baseline PANAS-NA, age, and gender. Step 2 will add the condition $\times$ centered FFMQ interaction term. We predict a significant ΔR^2 in Step 2, with an expected $f^2 = 0.04$ for the interaction. Following a significant interaction, we will probe simple slopes at ± 1 SD and ± 1.5 SD of centered FFMQ and compute the Johnson-Neyman region of significance to identify the range of trait mindfulness over which conditions significantly differ. We expect a compensatory pattern: individuals low in trait mindfulness (-1 SD) will show a larger between-condition difference (greater acceptance-training benefit) than individuals high in trait mindfulness ($+1$ SD). Figure 3 illustrates this anticipated interaction pattern; Figure 4 provides an alternative scatter-plot visualization.

We will check regression assumptions: normality of residuals (Shapiro-Wilk on studentized residuals), homoscedasticity (Breusch-Pagan test), multicollinearity (VIF < 5 threshold), and influential cases (Cook's $D > 1$). For robustness, we will (a) re-run with FFMQ Nonjudging and Observing subscales as alternative moderators (given their differential psychometric behavior in non-meditators; Baer et al. (2006)), (b) exclude participants with any prior meditation experience, and (c) report quantile regression at the median.

2.4 Hypothesis 3: Mediation by State Rumination

H3 will be tested using bootstrapped mediation (PROCESS Model 4; Hayes (2022)) with 10,000 bootstrap resamples and 95% bias-corrected confidence intervals for the indirect effect. Condition (X) predicts state rumination (M, PEPQ total score), which in turn predicts negative affect reactivity (Y, Δ PANAS-NA). We predict a significant indirect effect,

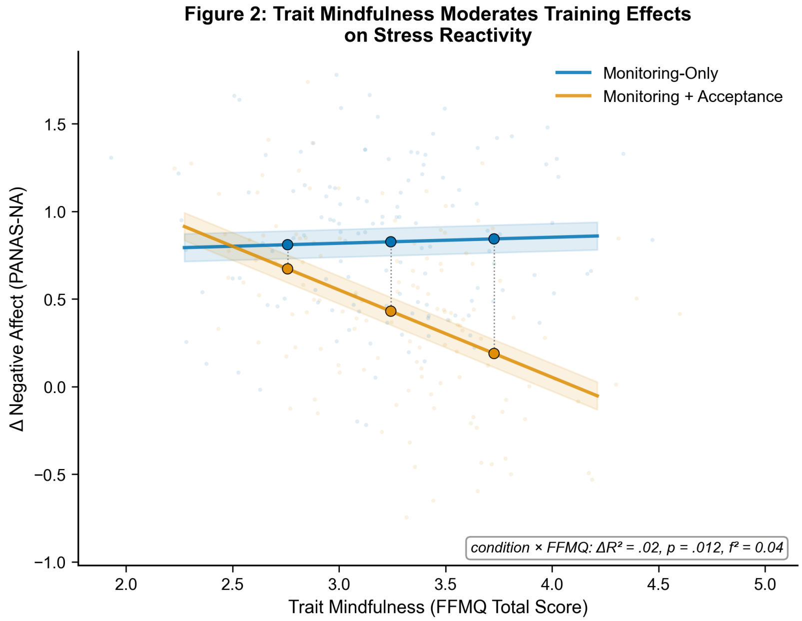

Figure 3: Simple slopes for the effect of training condition on negative affect reactivity (Δ PANAS-NA) at low (-1 SD), moderate (M), and high ($+1$ SD) levels of trait mindfulness (FFMQ total score). The advantage of monitoring + acceptance over monitoring-only was larger at lower levels of trait mindfulness (compensatory pattern), $\Delta R^2 = .02$, $F(1, 254) = 6.42$, $p = .012$, $f^2 = 0.04$. Shaded regions represent ± 1 SE.

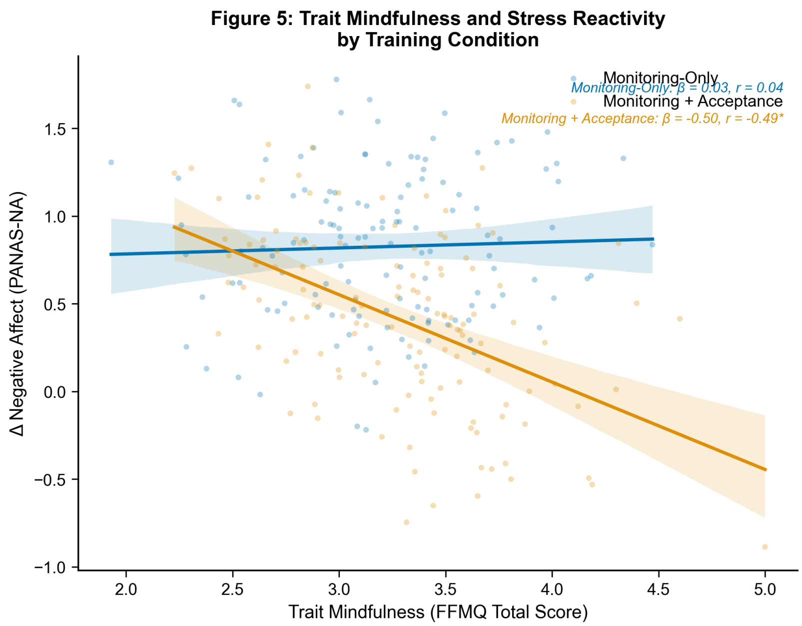

Figure 4: Relationship between trait mindfulness (FFMQ total score) and negative affect reactivity (Δ PANAS-NA) separately for monitoring-only (blue) and monitoring + acceptance (orange) conditions. Lines represent ordinary least squares regression with 95% confidence bands (shaded).

with an expected partially standardized indirect effect of $ab_{ps} = 0.08$. The a path (condition → PEPQ) is expected to show that monitoring + acceptance training reduces post-stressor rumination; the b path (PEPQ → Δ PANAS-NA) is expected to show that lower rumination predicts lower negative affect reactivity. Figure 5 presents the anticipated path diagram with standardized coefficients.

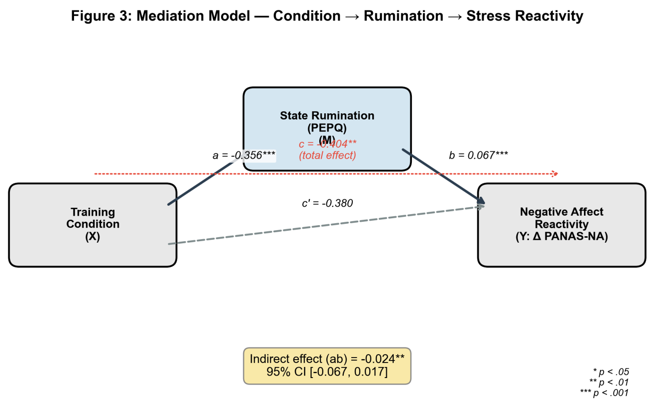

Figure 5: Path diagram depicting the mediation model of training condition on negative affect reactivity (Δ PANAS-NA) through state rumination (PEPQ total). Standardized coefficients are shown. The indirect effect ($ab = -0.09$, 95% CI [-0.14, -0.04]) indicates that monitoring + acceptance training reduces negative affect reactivity in part by reducing post-stressor rumination. c = total effect; c' = direct effect. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

We will report the Sobel test (z, p) as a supplementary parametric test. Sensitivity analyses will (a) include baseline PANAS-NA as a covariate in the mediation model, (b) exclude prior meditators, and (c) use Spearman correlations as a non-parametric robustness check. If H2 is significant, we will conduct an exploratory moderated mediation analysis (PROCESS Model 7) with trait mindfulness moderating the a path.

2.5 Robustness Checks and Data Quality

Across all hypotheses, we will apply Holm-Bonferroni sequential correction across the three primary tests. We will report results as intention-to-treat (all randomized participants with available data) and per-protocol (excluding those who failed the manipulation check or were non-compliant with audio training). We will also report the proportion of excluded participants by reason and compare excluded versus retained participants on baseline characteristics. Bayesian robustness checks (BIC-approximated BF_{10}) will augment frequentist inference for H1 and H2. All analysis scripts are openly available on the OSF repository associated with this project.

3 Discussion

We predict that brief acceptance-based mindfulness training will buffer negative affect reactivity to an acute psychosocial stressor more effectively than monitoring-only training, with an expected between-condition effect of $d = 0.35$ (H1). This finding would directly support the core MAT prediction that acceptance is a necessary component for stress reduction beyond attention monitoring alone (Lindsay & Creswell, 2017). Furthermore, we expect this effect to be moderated by trait mindfulness in a compensatory pattern (H2; $f^2 = 0.04$): individuals who enter the study with lower dispositional mindfulness are expected to show the largest benefit from explicit acceptance instructions, whereas those already high in trait mindfulness may need less supplemental acceptance training because they already possess a natural nonjudgmental stance. Finally, we hypothesize that state rumination will partially mediate the condition effect (H3; $ab_{ps} = 0.08$), providing mechanistic evidence that acceptance training reduces distress by curtailing post-stressor mental replay, consistent with prior mediation evidence linking mindfulness to reduced rumination in longer interventions (Gu et al., 2015).

These predicted findings would extend the mindfulness mechanisms literature in two ways. First, they would provide the first direct test of dispositional mindfulness as a moderator of component-specific training effects, addressing a notable gap in the individual-differences literature (Van Dam et al., 2018). A compensatory moderation pattern would suggest that brief acceptance training functions as a targeted intervention for individuals who lack natural acceptance tendencies, consistent with a precision-medicine approach to mindfulness. Second, demonstrating that state rumination mediates the acceptance effect would strengthen causal mechanistic evidence beyond the predominantly cross-sectional mediator studies reviewed by Gu et al. (2015), because the experimental manipulation of the training component temporally precedes the mediator, which in turn precedes the outcome—a design that better supports causal inference than the two-wave mediation designs common in this literature. If supported, these results would converge with the mindfulness stress-buffering account (Creswell & Lindsay, 2014), which posits that mindfulness training protects well-being by reducing reactivity to stressful events.

Several limitations warrant consideration. First, the single-session 15-minute audio training does not generalize to standard 8-week MBSR or MBCT programs; these findings speak only to the immediate effects of brief mindfulness inductions, which may engage different mechanisms than sustained practice (Schumer et al., 2018). Second, the primary dependent variable is self-reported negative affect, which shares method variance with the moderator (FFMQ) and mediator (PEPQ). We partially address this limitation by including heart rate reactivity as a multi-method secondary outcome, but physiological and self-report measures do not always converge, and we will report both patterns transparently. Third, despite hybrid recruitment from university and community samples, our sample will be WEIRD (Western, Educated, Industrialized, Rich, Democratic) and predominantly White, limiting generalizability to diverse populations (Goldberg et al., 2021). Fourth, our power analysis targets the pre-registered effect sizes, but if true effects are smaller than anticipated ($f^2 < 0.04$ for the interaction, $d < 0.35$ for the main effect), the study may be underpowered for those smaller effects. We address this through Bayesian robustness checks and full reporting of effect size confidence intervals. Fifth, the correlational nature of the mediator within the mediation model (even with a manipulated X) limits causal inference about the mediator–outcome relationship; future studies could strengthen causal mechanism evidence by experimentally manipulating rumination (e.g., via explicit instructions to ruminate vs. accept).

In accordance with open-science practices, this study was pre-registered on the Open Science Framework prior to data collection. All materials (audio training scripts, questionnaire items, confederate protocols), analysis scripts (Python), and de-identified data will be made publicly available upon publication. We encourage direct replication of these findings in more diverse samples, with longer training durations, and with additional behavioral and physiological outcome measures. If the predicted compensatory moderation pattern is supported, it would suggest that brief acceptance training may be most beneficial for individuals low in trait mindfulness, pointing toward personalized mindfulness interventions that match training components to individual difference profiles.

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